

Michael Psenka

✉ psenka@eecs.berkeley.edu • michaelpsenka.io • [Google Scholar](#)

Education

UC Berkeley

Aug 2021 – Present

PhD in Electrical Engineering and Computer Science

Berkeley, CA

GPA: 4.0/4.0 major, 3.95 overall. Member of [Berkeley AI Research \(BAIR\)](#). Stand-in prof. for [CS 70](#) (350+ students).

Advised by [Aditi Krishnapriyan](#). Collaborators include [Yann LeCun](#), [Pieter Abbeel](#), [Yi Ma](#), and [Shankar Sastry](#).

Princeton University

Aug 2017 – May 2021

B.A. in Mathematics (minors: CS, Applied Math)

Princeton, NJ

GPA: 3.6/4.0. Awards: Peter A. Greenberg '77 Memorial Prize in Mathematics (**solved an open problem** in spectral geometry; published in *Journal of Spectral Theory*), Manfred Pyka Memorial Prize in Physics, 1st place HackPrinceton.

About me: **Reinforcement learning, world models, AI+bio/chem**. Prev. pure mathematics researcher. Focus on longer-horizon and out-of-distribution planning. Multi-disciplinary, searching for the right unifying ideas to engineer intelligence.

Experience

Research Scientist Intern @ Meta FAIR (with [Yann LeCun](#))

May 2025 – Dec 2025

- Identified a failure mode in world-model planners and derived a new robust, parallelized, gradient-based planner.
- Achieved SOTA long-horizon planning performance in action-conditioned video models, improving success rates by up to $2\times$ and reducing planning time by up to $6\times$ versus prior methods.
- Led the project end-to-end as sole first author under Yann LeCun, culminating in an ICML 2026 paper under review, arXiv preprint, open-source release, and external adoption to other world-model repositories.

Undergraduate Researcher @ Princeton, Stanford

Jun – Sep 2019, 2020

- (Princeton) Developed SOTA second-order Riemannian optimization algorithm for tensor-train manifolds, contributing the implementation to the Manopt toolbox. Presented at NeurIPS OPT 2020.
- (Stanford) Developed a multi-view 3D reconstruction method that completely bypasses pairwise registration using non-Euclidean ML. Presented at IROS Geometric Methods Workshop 2020.

Machine Learning & AI Engineer @ Moovila, Inc.

Jan – Apr 2025, Jun – Aug 2018, '17, '16

- Co-inventor on a patent for mathematical modeling of large-scale project connections; remains a core component of the primary product delivered to all customers.
- Co-developed a custom LLM-integrated chatbot with tooling for improved product functionality and ease of use.

Selected Publications

- Parallel Stochastic Gradient-Based Planning for World Models.**
[M. Psenka](#), [M. Rabbat](#), [A. Krishnapriyan](#), [Y. LeCun*](#), [A. Bar*](#). *ICML 2026*. [\[paper\]](#) [\[code\]](#)
- Action-Minimization Meets Generative Modeling: Efficient Transition Path Sampling with the Onsager-Machlup Functional.**
[S. Raja*](#), [M. Šípká*](#), [M. Psenka*](#), [T. Kreiman](#), [M. Pavelka](#), [A. S. Krishnapriyan](#). *ICML 2025*. [\[paper\]](#) [\[code\]](#)
- Learning a Diffusion Model Policy from Rewards via Q-Score Matching.**
[M. Psenka*](#), [A. Escontrela*](#), [P. Abbeel](#), [Y. Ma](#). *ICML 2024*. [\[paper\]](#) [\[code\]](#)
- Representation Learning via Manifold Flattening and Reconstruction.**
[M. Psenka](#), [D. Pai](#), [V. Raman](#), [S. Sastry](#), [Y. Ma](#). *JMLR*, 2024. [\[paper\]](#) [\[code\]](#)
- Pursuit of a Discriminative Representation for Multiple Subspaces via Sequential Games.**
[D. Pai](#), [M. Psenka](#), [C.-Y. Chiu](#), [M. Wu](#), [E. Dobriban](#), [Y. Ma](#). *Journal of the Franklin Institute*, 2023. [\[paper\]](#) [\[code\]](#)
- A Proof of the Triangular Ashbaugh-Benguria-Payne-Pólya-Weinberger Inequality.**
[R. Arbon*](#), [M. Mannan*](#), [M. Psenka*](#), [S. Ragavan*](#). *Journal of Spectral Theory*, 2022. [\[paper\]](#)
- CTRL: Closed-Loop Transcription to an LDR via Minimizing Rate Reduction.**
[X. Dai](#), [S. Tong](#), [M. Li](#), [Z. Wu](#), [M. Psenka](#), [K. H. R. Chan](#), [P. Zhai](#), [Y. Yu](#), [X. Yuan](#), [H. Y. Shum](#), [Y. Ma](#). *Entropy*, 2022. [\[paper\]](#) [\[code\]](#)
- Second-Order Optimization for Tensors with Fixed Tensor-Train Rank.**
[M. Psenka](#), [N. Boumal](#). *NeurIPS OPT Workshop*, 2020. [\[paper\]](#) [\[code\]](#)
- Reconstruction Without Registration.**
[M. Psenka](#), [T. Birdal](#), [L. Guibas](#). *IROS Geometric Methods Workshop*, 2020. [\[paper\]](#) [\[code\]](#)

Skills

Research	RL, world models, diffusion models, stochastic interpolants, control theory, differential geometry
Tools	PyTorch, JAX, Python, C, C#, Linux, Git, Shell, AWS

Interests

Piano, snowboarding, boxing, skateboarding, pole vaulting, music production, chess, mac and cheese, whiskey